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Mechanical

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Electromagnetic energy transfer

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (1)$$

$\varrho = \sigma \delta =$ charge density in interior

$\sigma =$ charge density at border (surface)

$\delta =$ Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} \hat{=}$ electric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (2)$$

Thermal

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Ralativistic

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